

Digital Signal Analysis and Processing Exam Notes

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

5. FIR Filter Design

6 hrs · 70 questions across the 45 papers · worth about 12 of 80 marks · 541 marks, 14.8% of the archive · 3 syllabus sub-topics · 4th of the 8 chapters by weight.

Read this first:

- Three topics, and they divide cleanly by *what you have to do*:
 - Window method** TOP 28/45: a design calculation. Pick a window, get a length, truncate the ideal response.
 - Kaiser window** TOP 19/45: the same calculation with two formulas bolted on, for β and for the length.
 - Remez exchange** TOP 23/45: **pure bookwork**. 23 questions, almost always the same flowchart and the same three sentences, and **not one of them asks you to run the algorithm**.
- So a third of this chapter's marks are free if you memorise one flowchart. Do §5.6 first.
- The whole design method is two lines**: the ideal filter has an infinite, non-causal impulse response, so **truncate it** and **taper the truncation** so the edges do not ring.
- The window is chosen by **stopband attenuation** and the length by **transition width**. Those are two independent lookups and they are the only two numbers a fixed-window question turns on.
- Gibbs ripple does not shrink when you lengthen the filter**. That single fact is worth 3 to 6 marks in eight papers, and it is the reason Kaiser exists.

5.1 The window method

Describe how a digital FIR filter is designed by the window method, and discuss the characteristics of the different window functions 4+4 5+3 4 5

TOP 28/45 70 Ch 4+4, 72 Ch 5+3, 72 Ma 4, 70 Bh 5 · the answer is the four steps below, then the table in §5.2

Where the ideal response comes from

An ideal low-pass filter is a brick wall in frequency. Inverse DTFT it:

$$h_d[n] = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega = \frac{\sin \omega_c n}{\pi n}$$

- This is a sinc: **infinitely long** and **non-causal**, so it cannot be built.
- Delay it by $\alpha = \frac{N-1}{2}$ samples to make it causal, which is also exactly what gives **linear phase**:

$$h_d[n] = \frac{\sin(\omega_c(n-\alpha))}{\pi(n-\alpha)}, \quad h_d[\alpha] = \frac{\omega_c}{\pi}$$

- The centre tap is a 0/0 limit, so **evaluate it separately as ω_c/π** . Forgetting that is the most common slip in the chapter.

Why multiplying by a window is convolving in frequency

- $h[n] = h_d[n]w[n]$ becomes, in frequency,

$$H(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\theta}) W(e^{j(\omega-\theta)}) d\theta$$

The four steps, which are the whole method

- Write $h_d[n]$ for the ideal filter, delayed by $\alpha = (N-1)/2$.
- Choose the window** from the required stopband attenuation.
- Choose N** from the required transition width.
- Multiply: $h[n] = h_d[n]w[n]$ for $0 \leq n \leq N-1$, zero elsewhere.

Other ideal responses

- High pass**: $\delta[n-\alpha] - h_{d,LP}[n]$, an all-pass minus the low pass.
- Band pass**: $h_d(\omega_2) - h_d(\omega_1)$, the wider low pass minus the narrower.
- Every one of them is symmetric about α** , so every windowed FIR filter here has linear phase automatically.

- So the brick wall is **smeared by the window's spectrum**. The result is the sharp edge replaced by a transition, plus ripple from the window's sidelobes.
- **Main lobe width sets the transition; sidelobe height sets the ripple**. That one sentence explains the entire comparison table in §5.2.

5.2 The windows, and the Gibbs phenomenon

Explain the Gibbs phenomenon: how it arises when the rectangular window is used, and how it can be minimized 3+3 5 6 1+4 2+5

TOP 8/45 81 Ba, 80 Ba, 73 Shr, 71 Shr, 79 Ch, 74 Bh, 71 Bh, 81 Ch · asked in eight papers, always the same answer

What the Gibbs phenomenon is

- Truncating the sinc with a **rectangular** window convolves H_d with the Dirichlet kernel, whose sidelobes are only **13 dB** below its main lobe.
- The result overshoots the band edge by about **9 % of the discontinuity height**, with ringing either side.
- **Increasing N does not help**. The ripples get narrower and more numerous, but the overshoot stays at 9 %. The series converges in the mean-square sense, **not uniformly**.
- That is the whole phenomenon, and the fact examiners want stated.

How to minimize it

- **Use a tapered window**. Hann, Hamming and Blackman fall smoothly to zero at the ends instead of being cut off, so their sidelobes are far lower and the ripple is far smaller.
- The price is a **wider main lobe**, hence a wider transition band, hence a longer filter for the same sharpness.
- **Every window is a trade of ripple against transition width**.

The window formulas, for $0 \leq n \leq M$ with $M = N - 1$

$$w_{\text{rect}}[n] = 1, \quad w_{\text{Bart}}[n] = 1 - \frac{|2n - M|}{M}$$

$$w_{\text{Hann}}[n] = 0.5 - 0.5 \cos \frac{2\pi n}{M}, \quad w_{\text{Hamm}}[n] = 0.54 - 0.46 \cos \frac{2\pi n}{M}$$

$$w_{\text{Black}}[n] = 0.42 - 0.5 \cos \frac{2\pi n}{M} + 0.08 \cos \frac{4\pi n}{M}$$

- **Hann ends at exactly 0; Hamming ends at 0.08**. That is the only difference between them and it buys 9 dB of stopband.
- All five are **symmetric**, which is what keeps the phase linear.

The comparison table (learn this: it answers half the chapter)

Window	Main lobe	Peak side lobe	Stop band	Transition
Rectangular	$\frac{4\pi}{M+1}$	−13 dB	21 dB	$1.8\pi/M$
Bartlett	$8\pi/M$	−25 dB	25 dB	$6.1\pi/M$
Hann	$8\pi/M$	−31 dB	44 dB	$6.2\pi/M$
Hamming	$8\pi/M$	−41 dB	53 dB	$6.6\pi/M$
Blackman	$12\pi/M$	−57 dB	74 dB	$11\pi/M$

- **Stopband attenuation is a property of the window alone**. It does not depend on N at all, which is exactly why Gibbs ripple cannot be lengthened away.
- $M = N - 1$ throughout. Papers sometimes call M the length; read which.
- The two columns you actually use are **Stop band** (to choose the window) and **Transition** (to choose N).

5.3 Choosing the window and the length

The two lookups

- Window** ← stopband attenuation. Take the *cheapest* window whose figure meets the spec. If the paper gives a stopband ripple δ_s instead, convert first:

$$A_s = -20 \log_{10} \delta_s \text{ dB}$$

- Length** ← transition width $\Delta\omega = |\omega_s - \omega_p|$, from the same table:

$$N - 1 = M = \frac{c \pi}{\Delta\omega}$$

with c the window's transition constant (6.6 for Hamming, and so on).

- Round N **up**, and up again to the next **odd** number so $\alpha = (N - 1)/2$ is a whole sample.

The cut-off goes in the middle

- The window method has no independent control of the band edges, so put the ideal cut-off **halfway across the transition band**:

$$\omega_c = \frac{\omega_p + \omega_s}{2}$$

- Use ω_c , not ω_p , in $h_d[n]$.** Using the passband edge is a guaranteed lost mark.
- If the spec is in Hz, convert first: $\omega = 2\pi f/f_s$. If it is in rad/s, $\omega = \Omega/f_s$ (equivalently ΩT).

The window choice is a step, not a slope. 67 Mng asks for 54 dB.

Hamming gives 53, one short, so Blackman is forced and the filter jumps from 67 taps to 111 for the sake of 1 dB. Say so: it is the point of the question.

5.4 The Kaiser window

Why is the Kaiser window better than other fixed windows? Explain the steps to design an FIR filter using it 3 4 7 2+2

TOP 19/45 81 Ba, 74 Ash, 72 Ch, 80 Ch, 72 Ash, 74 Bh 2+2 · the "why" is one sentence, the design is four lines

Why it is better (the whole answer)

- Every fixed window offers **one** stopband attenuation, take it or leave it. Hann gives 44 dB whether you want 30 or 43.
- Kaiser has a **shape parameter** β , so the attenuation is *continuously adjustable*.
- So the two specifications **decouple**: β **buys attenuation**, N **buys transition width**, and neither is wasted on the other.
- The result is **the shortest filter** that meets a given spec among all the window methods, because nothing is over-designed.
- It is also very nearly optimal in the energy sense: it maximises the fraction of the window's energy in the main lobe.

The design equations

$$w[n] = \frac{I_0\left(\beta \sqrt{1 - \left(\frac{2n}{M} - 1\right)^2}\right)}{I_0(\beta)}, \quad 0 \leq n \leq M$$

- $\delta = \min(\delta_p, \delta_s)$, then $A = -20 \log_{10} \delta$ dB.

$$\beta = \begin{cases} 0 & A \leq 21 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21) & 21 < A \leq 50 \\ 0.1102(A - 8.7) & A > 50 \end{cases}$$

$$M = \frac{A - 8}{2.285 \Delta\omega}, \quad \text{length} = M + 1$$

- I_0 is the **modified Bessel function of the first kind**, order

zero: $I_0(x) = \sum_{k \geq 0} \left[\frac{(x/2)^k}{k!} \right]^2$, and $I_0(0) = 1$.

$\beta = 0$ makes the Kaiser window rectangular, and larger β tapers it harder.

The same δ gives the same β , so most papers in §2 of the companion share one value.

5.5 Optimum filters and the Remez exchange algorithm

What is an optimum filter? Explain the Remez exchange algorithm for FIR filter design, with its flowchart. Why is it superior to windowing? 1+5 2+6 4 9 3+10

TOP 23/45 75 Ch, 72 Ch, 70 Ch, 69 Ch, 73 Ch, 78 Bh, 76 Ch, 82 Bh, 81 Bh, 76 Ash, 73 Shr, 71 Shr, 79 Bh, 79 Ba, 82 Ba, 77 Ch, 74 Bh, 78 Ch, 80 Ch, 74 Ma, 72 Ma, 70 Bh, 69 Bh · **23 questions, 175 marks, and no calculation in any of them**

What "optimum" means

- The window method accepts whatever error the window happens to give, and that error is **largest at the band edges** and small in the middle. Most of the allowed tolerance goes unused.
- An **optimum** (equiripple, minimax, Parks-McClellan) filter instead spreads the error evenly, so the worst case is as small as possible:

$$\min_h \max_{\omega \in S} |E(\omega)|, \quad E(\omega) = W(\omega) [H_d(\omega) - H(\omega)]$$

- $W(\omega)$ is a **weight**, which is how passband and stopband ripple can be traded: set $W = \delta_s / \delta_p$ in the passband and 1 in the stopband.
- S is the bands of interest only, so **the transition band is simply left out** rather than being forced to behave.

The alternation theorem, which is why it works

- H is the unique best approximation **if and only if** $E(\omega)$ has at least **$r + 2$ alternations**: extrema of equal magnitude and alternating sign, where r is the number of free coefficients.
- So the answer is recognisable when you see it: **count the alternations**.
- That is what makes the filter *equiripple*: every ripple in a band reaches exactly the same height.

Why it beats windowing

- **Shorter filter** for the same specification. **The local worked comparison is worth quoting**: for $\omega_p = 0.4\pi$, $\omega_s = 0.6\pi$ and $\delta = 0.0116$, the equiripple design needs $M = 26$ while a Kaiser window at the same δ needs $M = 38$. **A third fewer taps**.
- **Independent control** of passband and stopband ripple through W .
- Handles arbitrary multi-band shapes, not just the standard four.

The Remez exchange algorithm, as a flowchart (**this is the marks**)

- 1. Input** the band edges, the desired response H_d , the weights W , and the order.
- 2. Guess** an initial set of $r + 2$ extremal frequencies, usually spaced evenly over the bands.
- 3. Solve** for the deviation δ on that set. It comes out in closed form, with no matrix inversion:

$$\delta = \frac{\sum_k \gamma_k H_d(\omega_k)}{\sum_k (-1)^k \gamma_k / W(\omega_k)}$$

- 4. Interpolate** the trial response H through the points using Lagrange interpolation, and evaluate $E(\omega)$ on a dense grid.
- 5. Exchange**: find the new local extrema of $|E|$ and make them the new extremal set.
- 6. Test**: if the largest $|E|$ on the grid no longer exceeds $|\delta|$, the alternation theorem is satisfied, so **stop**. Otherwise go back to step 3.
- 7. Output** the impulse response from the converged frequency response.

Each pass can only increase $|\delta|$ towards its optimum, so the iteration always converges. **Draw it as a loop with the test at the bottom**, label the exchange step, and the flowchart marks are safe. No paper in the 45 asks for a numerical iteration.

5.6 Last-minute recall

Say these without thinking

- $h_d[n] = \frac{\sin(\omega_c(n - \alpha))}{\pi(n - \alpha)}$, centre tap ω_c/π , $\alpha = \frac{N-1}{2}$.
- $\omega_c = \frac{\omega_p + \omega_s}{2}$, and $\Delta\omega = |\omega_s - \omega_p|$.
- Window from **attenuation**: 21 / 25 / 44 / 53 / 74 dB.
- Length from **transition**: 1.8 / 6.1 / 6.2 / 6.6 / 11, all $\times \pi / \Delta\omega$.
- $A = -20 \log_{10} \delta$. Kaiser:
 $\beta = 0.5842(A - 21)^{0.4} + 0.07886(A - 21)$ for
 $21 < A \leq 50$, and $M = \frac{A - 8}{2.285 \Delta\omega}$.
- Gibbs overshoot is **9 %** and **does not depend on N** .
- Alternation theorem: $r + 2$ equal-magnitude alternating extrema.
- $\omega = 2\pi f / f_s$ for a spec in Hz.

Mistakes that cost marks

- Using ω_p instead of ω_c in $h_d[n]$.
- **Evaluating the centre tap as 0/0** instead of ω_c/π .
- Choosing the window from the transition width and the length from the attenuation. It is the other way round.
- Taking an even N , which puts α half way between samples.
- Saying Gibbs ripple can be fixed by a longer filter. **It cannot.**
- Quoting δ_p when the spec gives δ_s smaller: Kaiser uses $\min(\delta_p, \delta_s)$.
- Writing J_0 for the Bessel function in the Kaiser window. **It is I_0** , the *modified* one, and one paper mislabels it.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. $\rightarrow$ reads “leads to / therefore”, $+$ reads “and”, $=$ reads “is”. $\checkmark$ means the property holds, $\times$ that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months:** 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II).
Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 Bh is EX 753 Regular, 74 Ma is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.